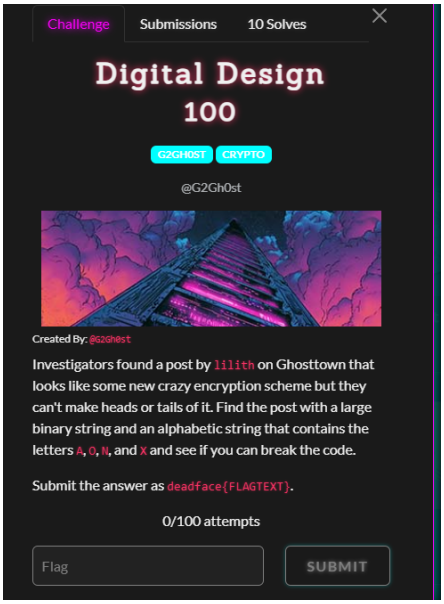
Solving “Digital Design” (100 pts - Crypto)

A Boolean-logic decoding walkthrough using AND, OR, NOT, and XOR operations.



Challenge page used during the solve.

1. The Challenge

A GhostTown post contained long binary strings paired with sequences made only from A, O, N, and X. The four letters are abbreviations for logic gates.

2. Map the Symbols

Symbol	Operation	Bits consumed
A	AND	2
O	OR	2
N	NOT	1
X	XOR	2

3. Evaluate Each Block

Process each five-letter gate sequence from left to right against its matching bit string. NOT consumes one input bit; the other gates consume two. Each sequence yields a five-bit output value.

```
Example:
Bits: 001100000
Gates: XXONO
Output: 00010
00010■ = 2 -> B
```

4. Convert the Outputs to Letters

Repeating the process across all blocks spells:

```
BINARYLOGIC
```

5. The Flag

```
deadface{BINARYLOGIC}
```

6. General Method

• Translate symbolic gate names before touching the binary.

• Track bit consumption carefully; NOT is unary while AND/OR/XOR are binary.

• Convert each five-bit result to an alphabet index and sanity-check the emerging English phrase.

Summary

Field	Result
Category	Cryptography / digital logic
Operations	AND, OR, NOT, XOR
Decoded text	BINARYLOGIC
Flag	deadface{BINARYLOGIC}